

# Active Directory

## Theory Notes

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Complete reference for IAM, IT Infrastructure, Cloud Security and SOC

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1. What is Active Directory? - Core concepts, architecture, why it exists
2. DNS - The Foundation - How clients find domain controllers
3. Kerberos Authentication - The protocol behind every login
4. LDAP and LDAPS - How apps query and talk to AD
5. Organisational Units and GPO - Structure and policy enforcement
6. Users, Groups and Permissions - IAM fundamentals - JML, AGDLP
7. AD Replication - How changes spread across DCs
8. FSMO Roles - The five special DC jobs
9. AD Sites and Services - Multi-site topology
10. PKI and AD CS - Certificates and LDAPS
11. Security Event IDs - What SOC analysts watch for
12. Attack Techniques - Spraying, Kerberoasting, DCSync and more
13. PowerShell for AD - Essential commands and scripts
14. Troubleshooting - How to diagnose AD problems

## Chapter 1

# What is Active Directory?

Core concepts, architecture, and why it exists

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## 1.1 The Problem AD Solves

Before Active Directory, every PC in a company had LOCAL accounts. A new hire meant creating accounts on EVERY machine separately. Changing a password meant doing it on every machine. Nightmare.

Active Directory solves this with a CENTRALISED DIRECTORY – one place to store identities, one place to manage access, one place to enforce policy.

## 1.2 Key Components

### Domain

The administrative boundary. Everything in labshandson.lan is ONE domain. Users, computers, policies all belong to it. Think of it as your company namespace.

### Domain Controller (DC)

A server running AD DS (Active Directory Domain Services). Stores the directory database (NTDS.dit), handles authentication, and runs DNS.

- DC01 – Forest root, first DC, holds all FSMO roles initially
- DC02 – Redundant DC, same site as DC01
- MUMDC01 – Branch DC in Mumbai, different subnet

### Forest

The TOP LEVEL container. A forest contains one or more domains that trust each other automatically. Your lab has ONE forest with ONE domain – the simplest and most common real-world design for small/medium companies.

### Global Catalog (GC)

A special DC that holds a PARTIAL copy of ALL objects in the entire forest – used to find objects across domains quickly. Every DC in a single-domain forest is automatically a GC.

## 1.3 The AD Database (NTDS.dit)

Active Directory stores everything in a file called NTDS.dit located at:

C:\Windows\NTDS\ntds.dit

This file contains ALL objects: users, computers, groups, GPOs, trusts. It is replicated between ALL Domain Controllers using a process called multi-master replication.

\* IMPORTANT: Never copy or move NTDS.dit while the DC is running. Use System State backup instead.

## 1.4 AD vs Workgroup

Workgroup: Each PC is independent. No central auth. No GPO. Max 10 computers recommended.

Active Directory: Centralised auth (Kerberos). Central policy (GPO). Scales to 100,000+ objects. Single login works everywhere.

## Chapter 2

# DNS – The Foundation

Without DNS, Active Directory does not work at all

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## 2.1 Why DNS is Critical for AD

AD uses DNS for EVERYTHING. When a client wants to log in, the FIRST thing it does is ask DNS: "Where is a Domain Controller I can talk to?" Without DNS working correctly, logins fail, GPO fails, replication fails.

\* Rule: If something is broken in AD, CHECK DNS FIRST.

## 2.2 SRV Records – The Secret Sauce

AD automatically creates special DNS records called SRV (Service) records when a DC is promoted. These tell clients WHERE to find AD services:

- \_ldap.\_tcp.labshandson.lan --> port 389 --> DC01
- \_kerberos.\_tcp.labshandson.lan --> port 88 --> DC01
- \_gc.\_tcp.labshandson.lan --> port 3268 --> DC01 (Global Catalog)

When a Windows client starts up, it queries DNS for these SRV records to FIND a DC. This is why DC01 must point its DNS to ITSELF (127.0.0.1).

## 2.3 AD-Integrated DNS Zones

In your lab, the DNS zone (labshandson.lan) is INTEGRATED into Active Directory - meaning the zone data is stored IN the AD database and replicates automatically between DCs. No separate zone transfer needed.

- Primary zone: Writable copy - your DCs all hold this
- Forwarders: Send unknown queries to 8.8.8.8 (Google) for internet resolution

## 2.4 DNS Record Types You Will See

A record: hostname --> IPv4 address

Win2K25-DC01.labshandson.lan --> 192.168.1.11

CNAME: alias pointing to another name

SRV: service location (how clients find DCs)

NS: nameserver records (who is authoritative for this zone)

SOA: Start of Authority - zone metadata, refresh intervals

## 2.5 DNS Troubleshooting Commands

nslookup labshandson.lan --> should return DC IP

nslookup -type=srv \_ldap.\_tcp.labshandson.lan --> SRV record check

dcdiag /test:dns --> full DNS health check

ipconfig /flushdns --> clear local DNS cache

ipconfig /registerdns --> re-register this machine in DNS

## Chapter 3

# Kerberos Authentication

The protocol behind every single login in your domain

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### 3.1 Why Kerberos? (Not NTLM)

NTLM (older protocol) sends password hashes over the network – vulnerable to relay attacks. Kerberos NEVER sends passwords. Instead it uses encrypted TICKETS that prove identity without transmitting credentials.

\* Kerberos is named after the three-headed dog guarding the underworld. The three heads = Client, KDC, and Service.

### 3.2 The Three Actors

- Client: The user/computer wanting access
- KDC (Key Distribution Center): Runs on EVERY DC. Has two parts – AS (Authentication Service) and TGS (Ticket Granting Service)
- Target Server: The resource being accessed (file server, app server etc.)

### 3.3 The Login Process – Step by Step

#### Step 1 – AS-REQ (Authentication Request)

Client sends request to KDC: "I am praveen.kumar, here is a timestamp encrypted with my password hash"

#### Step 2 – KDC Validates

KDC looks up praveen.kumar password hash in NTDS.dit. Decrypts the timestamp. If it works AND timestamp is within 5 minutes = identity confirmed.

#### Step 3 – AS-REP (TGT Issued)

KDC issues a TGT (Ticket Granting Ticket) – encrypted with the KRBTGT account hash. Client cannot read it but can use it to request service tickets later. Valid for 10 hours.

#### Step 4 – TGS-REQ (Service Ticket Request)

Client wants to access \\DC01\SalesDocs. Presents TGT to KDC and asks for a service ticket for the file server.

#### Step 5 – TGS-REP (Service Ticket Issued)

KDC validates TGT, issues a service ticket encrypted with the FILE SERVER account hash.

#### Step 6 – AP-REQ (Access!)

Client presents service ticket directly to the file server. Server decrypts it with its own hash, confirms it is valid. ACCESS GRANTED – without the KDC being involved again.

### 3.4 The 5-Minute Clock Rule

Kerberos requires all clocks to be within 5 minutes of each other. If clocks drift more than 5 minutes, authentication FAILS with a cryptic error. This is why w32tm (Windows Time Service) is critical and why the PDC Emulator FSMO role is the time master for the domain.

Check time sync: w32tm /query /status

### 3.5 Key Commands

klist --> list all cached Kerberos tickets

klist purge --> clear all tickets (forces re-auth)

setspn -L accountname --> list SPNs on an account

## Chapter 4

# LDAP and LDAPS

How applications query and communicate with Active Directory

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## 4.1 What is LDAP?

LDAP (Lightweight Directory Access Protocol) is the QUERY language for Active Directory. When an application needs to look up a user, check group membership, or search for computers, it uses LDAP.

- Port 389 - Plain LDAP (unencrypted - avoid in production)
- Port 636 - LDAPS (LDAP over SSL/TLS - use this in production)
- Port 3268 - Global Catalog LDAP (search entire forest)
- Port 3269 - Global Catalog LDAPS (encrypted)

## 4.2 LDAP vs Kerberos

Kerberos = AUTHENTICATION (proving who you are)

LDAP = QUERY (looking things up in the directory)

They work together: Kerberos proves your identity, LDAP searches the directory for information about you or other objects.

## 4.3 Distinguished Names (DN)

Every object in AD has a unique DN (Distinguished Name) that describes its FULL PATH in the directory tree:

CN=Praveen Kumar,OU=Users,OU=Delhi-HQ,DC=labshandson,DC=lan

- CN = Common Name (the object name)
- OU = Organisational Unit (the container)
- DC = Domain Component (each part of the domain name)

## 4.4 LDAPS - Why It Matters

Plain LDAP sends data unencrypted - usernames, attributes, and even password verification can be intercepted on the network. LDAPS wraps LDAP in TLS encryption, just like HTTPS wraps HTTP.

To enable LDAPS you need:

- A certificate issued to the DC (we used AD CS to do this)
- The certificate must have the DC FQDN as the subject
- DC must restart to bind the certificate to port 636

Test LDAPS: ldp.exe --> Connection --> Connect --> port 636 --> tick SSL --> Error 0 = success

## Chapter 5

# Organisational Units and Group Policy

Structuring the directory and enforcing settings

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## 5.1 Why OUs?

OUs (Organisational Units) are containers that serve TWO purposes:

- 1. ORGANISE objects logically (by department, location, function)
- 2. SCOPE Group Policy (GPOs apply to everything in an OU)

Your lab OU structure:

labshandson.lan

|-- Delhi-HQ

    |-- Users

    |-- Computers

    |-- Groups

|-- Mumbai-Branch

    |-- Users / Computers / Groups

|-- IT-Admins

## 5.2 GPO Processing Order - LSDOU

When multiple GPOs exist, they apply in this order (LAST one wins conflicts):

- L - Local Group Policy (on the machine itself)
- S - Site-linked GPOs
- D - Domain-linked GPOs (Default Domain Policy)
- O - OU-linked GPOs (closest OU wins)

\* Your Delhi-Password-Policy linked to Delhi-HQ OU wins over Default Domain Policy for Delhi users.

## 5.3 GPO Settings Structure

Every GPO has TWO configuration sections:

- Computer Configuration: applies when the COMPUTER starts up (password policy, firewall, software install)
- User Configuration: applies when the USER logs in (mapped drives, desktop settings, software)

Your password policy is under Computer Configuration because it controls the computer, not just the user.

## 5.4 Essential GPO Commands

gpupdate /force --> force immediate refresh

gpresult /r --> see what GPOs applied

gpresult /h report.html --> full HTML report

## 5.5 GPO Troubleshooting

GPO not applying? Check in this order:

- 1. Is the GPO linked to the correct OU?
- 2. Is the link enabled? (right-click the link in GPMC)
- 3. Is Security Filtering excluding the user/computer?
- 4. Is there a WMI filter blocking it?

- 5. Did gpupdate /force run successfully?
- 6. Check gpresult /h report - look at Denied GPOs section

## Chapter 6

# Users, Groups and Permissions

The IAM fundamentals - JML lifecycle and AGDLP

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## 6.1 The Joiner-Mover-Leaver (JML) Lifecycle

### Joiner - New Employee

- Create user account in correct OU
- Set password, tick "must change at next logon"
- Add to appropriate security groups
- Set properties: department, title, manager
- Verify access to required resources

### Mover - Role/Department Change

- Move account to new OU (if department OU design)
- Remove from old department groups
- Add to new department groups
- Update attributes: department, title, manager

### Leaver - Employee Leaving

- STEP 1: Disable account IMMEDIATELY (never delete first)
- STEP 2: Reset password to something unknown
- STEP 3: Remove from all security groups
- STEP 4: Move to Leavers/Disabled OU
- STEP 5: Delete after retention period (30-90 days typical)

\* Disable FIRST. If you delete immediately and a ticket comes in saying "actually they are staying" you cannot recover. Disabled accounts CAN be re-enabled.

## 6.2 Group Types and Scopes

### Security Groups

Used to assign permissions to resources. This is what you want 99% of the time.

### Distribution Groups

Email lists only. Cannot be used for permissions.

### Scopes

- Domain Local: assign permissions to resources in THIS domain. Members from ANY trusted domain.
- Global: group USERS from THIS domain. Can be used for permissions in any trusted domain.
- Universal: members from ANY domain in the forest. Used for forest-wide groups.

## 6.3 AGDLP Best Practice

AGDLP = Accounts --> Global groups --> Domain Local groups --> Permissions

Example:

praveen.kumar (A) is added to...

Delhi-Sales-Team Global group (G) which is added to...

SalesDocs-Modify Domain Local group (DL) which has...

Modify permission on SalesDocs share (P)

\* Benefit: To add a new member to the sales team, just add them to Delhi-Sales-Team. They automatically get the right permissions everywhere.

## 6.4 NTFS vs Share Permissions

Share permissions: control access via the NETWORK. Apply when accessing via UNC path (\\server\share)

NTFS permissions: control access at the FILE SYSTEM level. Apply regardless of how you access.

When BOTH apply: the MORE RESTRICTIVE permission wins.

Best practice: Set Share permissions to Full Control for the security group, then NTFS permissions to exactly what you want. All real security lives in NTFS.

## Chapter 7

# AD Replication

How changes spread across all Domain Controllers

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## 7.1 Multi-Master Replication

Unlike old databases with one master server, AD uses MULTI-MASTER replication. You can make changes on ANY domain controller (except for FSMO-protected operations) and those changes replicate to ALL other DCs.

## 7.2 How Replication Works – USN Numbers

Every change in AD gets a USN (Update Sequence Number). Each DC tracks the highest USN it has received from every other DC. When replication runs, DCs exchange USNs to figure out what has changed since the last sync.

- intra-site replication: between DCs in the SAME site – every 15 seconds
- inter-site replication: between DCs in DIFFERENT sites – scheduled (your lab: 180 minutes)

## 7.3 The KCC (Knowledge Consistency Checker)

The KCC is an automatic process that runs on every DC and calculates the most EFFICIENT replication topology. You do not need to manually configure who replicates with whom – the KCC figures it out automatically based on site links.

## 7.4 SYSVOL Replication

SYSVOL is a shared folder on every DC containing:

- Group Policy template files
- Logon scripts

SYSVOL replicates separately from the AD database using DFS-R (Distributed File System Replication). If SYSVOL replication breaks, GPOs stop applying.

## 7.5 Replication Commands

repadmin /replsummary --> quick health check (0 failures = good)

repadmin /showrepl --> detailed replication status

repadmin /syncall /A /e /P --> force replication now

dcdiag /test:replications --> formal replication test

## 7.6 Common Replication Error Codes

Error 58: RPC server unavailable – DC unreachable, firewall blocking ports, or DC offline

Error 8453: Replication access was denied – permission issue

Error 1722: The RPC server is unavailable – network/DNS issue

Error 8464: Replication in progress – wait and retry

## Chapter 8

# FSMO Roles

The five special jobs only one DC at a time can hold

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## 8.1 Why FSMO Roles?

Multi-master replication is great but some operations CANNOT safely happen on multiple DCs simultaneously. FSMO (Flexible Single Master Operations) roles designate ONE DC to handle these operations exclusively, preventing conflicts.

## 8.2 The Five Roles

### Schema Master (Forest-wide)

The ONLY DC that can modify the AD schema. The schema defines what object types and attributes can exist. Changed rarely - only when installing Exchange, Lync, or extending schema.

### Domain Naming Master (Forest-wide)

Controls adding and removing domains from the forest. Used when creating child domains or removing domains.

### RID Master (Domain-wide)

Allocates blocks of RIDs (Relative IDs) to other DCs. Every AD object has a SID = domain SID + RID. The RID master prevents duplicate SIDs.

### PDC Emulator (Domain-wide)

The BUSIEST FSMO role. Handles:

- Time synchronisation master for the domain
- Urgent password changes (password reset here first, syncs everywhere)
- Account lockout decisions
- Group Policy processing

### Infrastructure Master (Domain-wide)

Updates cross-domain object references. In a single-domain forest (like yours) this role is less important but still required.

## 8.3 Key Commands

netdom query fsmo --> see which DC holds each role

Transfer: use ADUC right-click --> Operations Masters (for RID/PDC/Infrastructure)

Transfer Schema Master: register schmmgmt.dll, then MMC snap-in

Transfer Domain Naming Master: Active Directory Domains and Trusts

## 8.4 Transfer vs Seize

Transfer: graceful handover while original holder is ONLINE. Always do this.

Seize: forceful takeover when original holder is PERMANENTLY DEAD. Use ntdsutil. Only as last resort - the original holder should NEVER come back online after a seize.

## Chapter 9

# AD Sites and Services

Managing multi-site topology and replication

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## 9.1 What is a Site?

An AD Site represents a physical location with FAST, RELIABLE network connectivity between its members. Sites tell AD "these computers are close together, these are far apart."

- Delhi-HQ site: 192.168.1.0/24 - DC01 and DC02
- Mumbai-Branch site: 192.168.10.0/24 - MUMDC01

## 9.2 Why Sites Matter

Without sites, AD assumes everything is one fast network and replicates constantly (every 15 seconds). Sites let you say "Mumbai is connected via a slower link" so replication is scheduled sensibly (every 180 min in our lab).

Sites also control which DC a client authenticates against - clients prefer DCs in their OWN site (faster, local connection).

## 9.3 Subnets

Subnets in AD Sites and Services tell AD which IP ranges belong to which site. When a client joins the domain, AD checks its IP against the subnet list to determine which site it belongs to.

Your subnets:

- 192.168.1.0/24 --> Delhi-HQ
- 192.168.10.0/24 --> Mumbai-Branch

## 9.4 Site Links

A site link defines the connection BETWEEN two sites. Key settings:

- Cost: lower = more preferred path (like routing metrics)
- Replication interval: how often to replicate (your lab: 180 min)
- Schedule: when replication is allowed

\* In a real WAN environment you would set cost and interval to match actual bandwidth - a 1Gbps fibre link gets low cost/frequent replication, a 4G backup link gets high cost/less frequent.

## Chapter 10

# PKI and AD CS

Certificates, encryption and trusted identity

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## 10.1 What is PKI?

PKI (Public Key Infrastructure) is a system of digital certificates, certificate authorities and cryptographic keys that enables:

- Encryption: scrambling data so only the intended recipient can read it
- Authentication: proving a server/user is who they claim to be
- Integrity: proving data has not been tampered with

## 10.2 How Certificates Work

A certificate is a digital document containing:

- The subject name (who it is for, e.g. Win2K25-DC01.labshandson.lan)
- The public key
- The issuer (who signed it - your CA)
- Validity dates
- Intended purposes (Server Authentication, Client Authentication etc.)

Anyone can VERIFY a certificate using the CA public key. If the CA signed it, the certificate is trusted.

## 10.3 AD CS (Active Directory Certificate Services)

AD CS lets you run your OWN Certificate Authority inside your domain. Benefits:

- Issue certificates to DCs automatically (auto-enrollment)
- Enable LDAPS without paying a public CA
- Issue smart card certificates for passwordless auth
- Issue certificates to internal web servers

### Your Lab CA

Enterprise Root CA: labshandson-WIN2K25-DC01-CA

Once installed, DC01 auto-enrolled for a Domain Controller certificate, which LDAP binds to port 636 enabling LDAPS.

## 10.4 Why LDAPS Matters for Security

Plain LDAP (port 389) sends all data unencrypted. An attacker on the same network can intercept:

- Usernames and attributes being queried
- Simple bind authentication credentials

LDAPS (port 636) encrypts ALL communication using TLS. Mandatory for any production environment or compliance requirement (SOC 2, ISO 27001 etc.)

## Chapter 11

# Security Event IDs

What SOC analysts watch for in AD environments

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## 11.1 Enabling Audit Policies

By default, Windows does NOT log everything. You must ENABLE audit policies first. Use auditpol or Group Policy:

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
```

```
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
```

## 11.2 Critical Event IDs

### Authentication Events

4624 - Successful logon

- Logon Type 2 = interactive (at keyboard)
- Logon Type 3 = network (accessing shared resource)
- Logon Type 10 = remote interactive (RDP)

4625 - Failed logon

- Sub Status 0xC000006A = wrong password (account exists)
- Sub Status 0xC0000064 = account does not exist
- Sub Status 0xC0000234 = account locked out

4740 - Account locked out

4767 - Account unlocked

### Account Management Events

4720 - User account created

4726 - User account deleted

4738 - User account changed

4781 - Account name changed

### Group Membership Events - CRITICAL for SOC

4728 - Member added to security-enabled GLOBAL group

4732 - Member added to security-enabled LOCAL group

4756 - Member added to security-enabled UNIVERSAL group

\* ALERT ON: Any addition to Domain Admins, Enterprise Admins, Schema Admins. These are high privilege groups that should rarely change.

### Kerberos Events

4768 - Kerberos TGT requested (AS-REQ/AS-REP)

4769 - Kerberos service ticket requested (TGS-REQ/TGS-REP)

4662 - Operation performed on an object (used to detect DCSync)

## 11.3 PowerShell Log Queries

```
Get-WinEvent -FilterHashtable @{LogName="Security"; Id=4625} -MaxEvents 10
```

```
Get-WinEvent -FilterHashtable @{LogName="Security"; Id=4728} -MaxEvents 10
```

## Chapter 12

# Attack Techniques

Understanding how AD gets compromised - to defend it better

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## 12.1 Password Spraying

### How it works

Attacker tries ONE common password (e.g. Summer2026!) against HUNDREDS of accounts. Stays under per-account lockout threshold (e.g. tries only 1-2 attempts per account).

### Detection

Event ID 4625 - many failures across DIFFERENT accounts from the SAME source IP in a short window. Normal brute force = many failures on ONE account.

### Defence

- Strong password policy (your GPO: 12 char minimum)
- MFA (makes spraying useless even if password found)
- Account lockout after 5 attempts

## 12.2 Kerberoasting

### How it works

Any authenticated domain user can request a Kerberos Service Ticket for any account with an SPN (Service Principal Name). The ticket is encrypted with the SERVICE ACCOUNT password hash. Attacker cracks it offline.

### Detection

Event ID 4769 - high volume of service ticket requests from one account in short time. Unusual service names being requested.

### Defence

- Long random passwords on all service accounts (20+ chars)
- Use gMSA (Group Managed Service Accounts) - passwords rotate automatically

## 12.3 DCSync

### How it works

An attacker with replication rights (Domain Admin or specifically granted "Replicating Directory Changes") uses AD own replication protocol to request password hashes for ALL accounts directly from a DC. No malware on the DC needed.

### Detection

Event ID 4662 with specific replication GUIDs. An account performing replication that is NOT a DC account is suspicious.

### Defence

- Restrict "Replicating Directory Changes" permission to only actual DCs
- Monitor for 4662 from non-DC accounts

## 12.4 Pass-the-Hash

### How it works

Attacker extracts NTLM password hash from memory of a compromised machine (using mimikatz or similar). Uses the HASH directly to authenticate - never needs the plaintext password.

### Defence

- Credential Guard (Windows Defender feature)
- LAPS (Local Administrator Password Solution) - unique local admin passwords
- Tiered admin model - admin accounts never log into workstations

## 12.5 Golden Ticket

### How it works

If an attacker gets the KRBTGT account hash (from DCSync or compromising a DC), they can FORGE Kerberos TGTs for any user, any group membership, any privilege - valid indefinitely.

### Why it is the worst AD attack

A Golden Ticket works EVEN AFTER the compromised user password is changed. The only defence is rotating krbtgt password TWICE (because of KDC caching) and monitoring for anomalous Kerberos tickets.

## Chapter 13

# PowerShell for AD

Essential commands for daily administration and automation

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## 13.1 Getting Started

Import-Module ActiveDirectory --> load the AD module

Get-Command -Module ActiveDirectory --> see all available cmdlets

## 13.2 User Commands

Get-ADUser -Filter \* | Select Name, SamAccountName, Enabled

Get-ADUser -Identity praveen.kumar -Properties \* | Select Name, LastLogonDate, Department

New-ADUser -Name "Name" -SamAccountName logon -Path "OU=..." -Enabled \$true -AccountPassword  
(ConvertTo-SecureString "Pass" -AsPlainText -Force)

Set-ADUser -Identity username -Department "IT" -Title "Analyst"

Disable-ADAccount -Identity username

Enable-ADAccount -Identity username

Unlock-ADAccount -Identity username

Set-ADAccountPassword -Identity username -NewPassword (...) -Reset

Remove-ADUser -Identity username -Confirm:\$false

## 13.3 Group Commands

Get-ADGroup -Filter \* | Select Name, GroupScope, GroupCategory

Get-ADGroupMember -Identity "GroupName" | Select Name

Add-ADGroupMember -Identity "Group" -Members username

Remove-ADGroupMember -Identity "Group" -Members username -Confirm:\$false

## 13.4 Audit and Security Scripts

-- Stale accounts (not logged in 90+ days) --

\$cutoff = (Get-Date).AddDays(-90)

Get-ADUser -Filter {LastLogonTimestamp -lt \$cutoff -and Enabled -eq \$true} -Properties LastLogonTimestamp |  
Select Name, @{N="LastLogon";E=[[DateTime]::FromFileTime(\$\_.LastLogonTimestamp)]}

-- Password never expires (security risk) --

Get-ADUser -Filter {PasswordNeverExpires -eq \$true -and Enabled -eq \$true} | Select Name, SamAccountName

-- Who is in Domain Admins? --

Get-ADGroupMember -Identity "Domain Admins" | Select Name, SamAccountName

## 13.5 Bulk Operations

-- Import from CSV and create users --

Import-Csv "C:\users.csv" | ForEach-Object {

New-ADUser -Name \$\_.Name -SamAccountName \$\_.Sam ...

}

-- Export all users for access review --

Get-ADUser -Filter \* -Properties Department, LastLogonDate | Export-Csv "C:\audit.csv" -NoTypeInfo

## Chapter 14

# Troubleshooting

Diagnosing and fixing common AD problems

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## 14.1 The Troubleshooting Mindset

Always work from the BOTTOM UP:

- 1. Can you PING the target? (Network layer)
- 2. Can you RESOLVE the name in DNS? (DNS layer)
- 3. Can you reach the service port? (Service layer)
- 4. Does authentication work? (Auth layer)
- 5. Is the application/GPO behaving correctly? (App layer)

## 14.2 Essential Diagnostic Commands

dediag /v --> comprehensive DC health check

repadmin /replsummary --> quick replication check

repadmin /showrepl --> detailed replication + error codes

netdom query fsmo --> who holds FSMO roles

nltest /dsgetdc:domain --> can a DC be found?

gpresult /r --> what GPOs applied and why

w32tm /query /status --> time sync status

klist --> cached Kerberos tickets

## 14.3 Common Problems and Fixes

### "Trust relationship failed"

Machine account out of sync. Fix: Test-ComputerSecureChannel -Repair (run on affected machine)

### "Domain controller not found"

DNS problem. Check: nslookup labshandson.lan from the affected machine. Verify DNS server is pointing to a working DC.

### GPO not applying

Run gpresult /h report.html and check Denied GPOs section. Most common causes: wrong OU, security filtering, empty GPO.

### Replication error 58

RPC unavailable. Usually means firewall blocking ports or DC is offline. Check: ping the DC, check Windows Firewall, check OPNsense rules.

### "Access is denied" on a share despite correct group

Kerberos ticket was issued before the group was added. User must log OFF and back ON to get a fresh ticket with updated group membership.

### LDAPS not working after CA install

Restart the DC. The LDAP service needs to restart to bind the new certificate to port 636.

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End of Notes – labshandson.lan

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